When Does Cross-Lingual Calibration Transfer Work? Geometric Analysis and Practical Solutions

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Abstract

Probing internal representations for correctness is a promising path toward calibrating large language models, yet nearly all prior work is English-only. We present the first large-scale study of cross-lingual correctness probe transfer, evaluating two MoE architectures (Qwen3-235B, Llama 4 Scout) across 20 languages spanning high-, mid-, and low-resource tiers on GlobalMMLU, Belebele, and MMLU-ProX. We report three practical findings and one systematic negative result. **(1)** Linear projection (Procrustes alignment) with $\sim$70 parallel sentence pairs reaches 95% of maximum transfer improvement, with resource-tier-dependent saturation ($\sim$35 pairs for high-resource, $\sim$75 for low-resource languages). **(2)** Procrustes alignment reverses its effect around a zero-shot AUROC threshold of $\sim$0.55: it helps misaligned languages but hurts already well-aligned ones, providing a practitioner decision rule. **(3)** Approximately 30% of low-resource language–layer probing results are capacity artifacts (selectivity < 0.05), a methodological warning for the cross-lingual probing field. As a **negative result**, CaCLR—a consistency loss based on MMD distributional matching—produces no significant AUROC improvement across a full β sweep (ΔAUROC$= +0.002$, $p = 0.42$). CKA subspace analysis explains this failure: MMD matches marginal probe-logit distributions but cannot align the low-dimensional correctness direction in activation space, while linear projection directly learns the subspace rotation. A Simpson's Paradox in CKA–transfer correlation provides architecture-dependent geometric insight: controlling for depth reveals a significant positive partial correlation on Llama 4 Scout ($\rho = 0.29$, $p = 0.018$) but only a trending relationship on Qwen3-235B ($\rho = 0.16$, $p = 0.072$).

1 Introduction

Multilingual large language models exhibit calibration quality that degrades sharply outside English: non-English languages suffer up to 5$\times$ worse Expected Calibration Error Joshi et al. [2024]. Output-based recalibration (e.g., temperature scaling) cannot address this because the verbalized confidence and internal correctness directions are near-orthogonal Miao and Ungar [2026], and per-language solutions such as LACE Zhou et al. [2024] require target-language labels that are unavailable for most of the world's languages.

Recent work has shown that probing internal representations for correctness is effective in English: CORAL Miao et al. [2026a] learns cross-task correctness probes, Miao & Ungar [2026] demonstrate that correctness directions can steer model outputs, and Confidence Manifold Ji et al. [2024] locates manipulable calibration directions. Separately, Xiao et al. [2025] show that *knowledge-boundary* probes (input-level answerability) transfer across 8 languages. However, *correctness* probes—which detect whether a specific output is right or wrong, a fundamentally different and output-dependent signal—have never been tested multilingually.

We present the first large-scale study of cross-lingual correctness probe transfer. Using two MoE architectures (Qwen3-235B-A22B and Llama 4 Scout 17B-16E) and 20 languages across three resource tiers, we probe every fourth layer and evaluate five transfer strategies: zero-shot, mean-shift, CaCLR (a consistency loss using MMD on probe logits from parallel inputs), linear projection, and few-shot fine-tuning. This systematic evaluation reveals when and why transfer succeeds or fails, and provides practical guidance for deploying cross-lingual calibration.

Our contributions are: **(1)** A depth–transferability Pareto frontier that is architecture-dependent: Qwen3 shows strong tradeoff (16/20 languages with negative depth–transfer correlation) while Llama 4 shows a weaker, language-specific pattern (13/20 negative); **(2)** A systematic negative result for CaCLR with geometric explanation—MMD distributional matching cannot align the low-dimensional correctness subspace, while linear projection with $\sim$70 parallel pairs provides an effective and sample-efficient solution; **(3)** A methodological warning that $\sim 30\%$ of low-resource probing results are capacity artifacts, with Procrustes effect reversal at zero-shot AUROC≈ 0.55 as a practitioner decision threshold.

2 Related Work

LLM Calibration. Joshi et al. [2024] establish that multilingual LLMs are poorly calibrated outside English. Temperature scaling Guo et al. [2017] addresses post-hoc recalibration but cannot resolve cross-lingual representation misalignment. LACE Zhou et al. [2024] proposes per-language layer ensembles but requires target-language labels.

Representation-Based Calibration. CORAL Miao et al. [2026a] trains cross-task correctness probes on English representations. Miao & Ungar [2026] discover that correctness and verbalized confidence occupy near-orthogonal directions (cosine < 0.04) and use this for inference-time steering. Ji et al. [2024] find manipulable calibration directions via the Confidence Manifold. All work exclusively in English.

Cross-Lingual Probing. Xiao et al. [2025] transfer knowledge-boundary probes across 8 languages, finding that answerability—a binary, input-level property—transfers reasonably. Correctness is fundamentally different: it depends on the specific output and involves language-specific capability components that may not share cross-lingual structure. Civelli et al. study cross-lingual difficulty probe transfer, which informs our decomposition analysis.

Cross-Lingual Alignment. MUSE Conneau et al. [2018] and VecMap Artetxe et al. [2018] demonstrate that linear projections effectively align cross-lingual word embeddings. Our linear projection is conceptually analogous—Procrustes alignment applied to sentence-level correctness representations—and we characterize its sample efficiency for this new application.

Domain Adaptation Theory. Ben-David et al. [2010] provide the H-divergence framework for transfer bounds. Zhao et al. [2019] prove that marginal distribution matching fails when conditional distributions $P(Y|X)$ differ across domains—directly relevant to our CaCLR negative result, where language-specific capability components create differing conditionals.

Probing Methodology. Hewitt & Liang [2019] introduce control tasks to measure probe selectivity; Voita & Titov [2020] propose MDL probes. We apply Hewitt & Liang's framework to cross-lingual transfer and discover widespread capacity artifacts in low-resource settings.

Representation Similarity. CKA Kornblith et al. [2019] and SVCCA Raghu et al. [2017] measure representational alignment. We use linear CKA to analyze cross-lingual representation divergence across depth and discover a Simpson's Paradox: controlling for depth reverses the sign of the CKA–transfer correlation.

Conformal Prediction. Conformal prediction Angelopoulos and Bates [2023] provides distribution-free coverage guarantees but requires per-language calibration data. Our approach targets zero-label transfer and is complementary.

3 Method

3.1 Cross-Lingual Correctness Probing

Given a multilingual LLM with L layers, we extract hidden-state activations $\mathbf{h}_l \in \mathbb{R}^d$ at layer l for each input. For English, binary correctness labels $y \in \{0, 1\}$ are available from GlobalMMLU evaluation. We train a logistic regression (LR) probe $f_l(\mathbf{h}_l) = \sigma(\mathbf{w}_l^\top \mathbf{h}_l + b_l)$ on English activations and evaluate transfer to target languages.

We choose LR over more expressive probes (e.g., 2-layer MLP) based on empirical evidence: while MLPs achieve $\sim$2pp higher English AUROC, they suffer $\sim$5pp larger transfer gaps due to fitting English-specific activation patterns (Section 4.9.2).

3.2 Transfer Methods

Zero-Shot Transfer. Apply the English-trained probe directly to target-language activations.

Mean-Shift. Subtract the mean activation difference between English and target-language representations before applying the probe. Under standardized logistic regression, this adds a constant offset to all logits; since AUROC is rank-invariant, $\Delta\text{AUROC} \equiv 0$ by construction.

CaCLR. We formulate Calibrated Cross-Lingual Regularization as a multi-objective loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{MMD}} \tag{1}$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy on English labels, $\mathcal{L}_{\text{SB-ECE}}$ is soft-binned ECE for calibration, and $\mathcal{L}_{\text{MMD}}$ is the Maximum Mean Discrepancy between English and target-language *probe logits* computed from parallel inputs (same GlobalMMLU items in both languages).

MMD uses a Gaussian RBF kernel with median-heuristic bandwidth:

$$\text{MMD}^2 = \frac{1}{n^2}\sum_{i,j} k(z_i^{\text{en}}, z_j^{\text{en}}) - \frac{2}{nm}\sum_{i,j} k(z_i^{\text{en}}, z_j^{\text{tgt}}) + \frac{1}{m^2}\sum_{i,j} k(z_i^{\text{tgt}}, z_j^{\text{tgt}}) \tag{2}$$

where $z = \mathbf{w}^\top \mathbf{h} + b$ are scalar probe logits. Critically, for linear probes, $z \in \mathbb{R}^1$—MMD operates in a 1-D space. This is a design limitation: MMD matches marginal distributions of scalar logits but cannot capture rotations in the underlying high-dimensional activation space.

Implementation. Optimizer: Adam, lr $= 10^{-3}$, cosine schedule. Epochs: 50, early stopping with patience 10 on validation BCE. Initialization: from standard LR solution. Batch size: 256. Parallel set sizes: 8,200–12,800 per language from GlobalMMLU train split (70/30 stratified split for train/eval).

Linear Projection (Procrustes). Given n parallel sentence pairs with activations $(\mathbf{H}^{\text{en}}, \mathbf{H}^{\text{tgt}}) \in \mathbb{R}^{n \times d}$, we learn an orthogonal rotation $\mathbf{W}^* = \arg\min_{\mathbf{W}^\top\mathbf{W}=\mathbf{I}} \|\mathbf{H}^{\text{en}} - \mathbf{H}^{\text{tgt}}\mathbf{W}\|_F^2$ via Procrustes alignment, then apply the English probe to $\mathbf{H}^{\text{tgt}}\mathbf{W}^*$. This directly learns the subspace rotation mapping target-language correctness directions to English.

Few-Shot Fine-Tuning. Retrain probe parameters on k labeled target-language examples ($k \in \{25, 100\}$), initialized from the English probe solution.

3.3 Difficulty–Capability Decomposition

To analyze what drives transfer degradation, we decompose the correctness signal. We train IRT difficulty probes on item-level difficulty parameters (language-agnostic), extract the difficulty direction, project it out of activations, and retrain correctness probes on the residual. If transfer improves after projection, the removed component was shared (helpful for transfer); if it degrades, the removed component was language-specific.

3.4 CKA Geometric Analysis

We compute linear CKA between English and target-language activations at each probed layer to track cross-lingual representation divergence with depth. To disentangle CKA from the depth confound, we compute partial Spearman correlations between CKA and transfer AUROC controlling for layer depth.

4 Experiments

4.1 Setup

Models. Qwen3-235B-A22B (94 layers, MoE, INT4 quantization; 24 probed layers, every 4th) and Llama 4 Scout (48 layers, 17B-16E MoE, W4A16; 12 probed layers). Quantization validation (Section 4.9.4) confirms negligible artifacts ($\max |\Delta\text{AUROC}| < 0.012$, CKA$> 0.97$ between quantized and FP16).

Languages. 20 languages in three tiers: HIGH (zh, es, fr, de, pt, ja), MID (ar, ru, ko, hi, vi, tr), LOW (sw, bn, th, te, yo, am).

Benchmarks. Primary: GlobalMMLU. Generalization: Belebele (reading comprehension), MMLU-ProX (translated MMLU).

Metrics. AUROC (discrimination), ECE (calibration). All comparisons include bootstrap 95% CIs (1,000 resamples) and paired permutation tests. Effect sizes reported with Cohen's d.

4.2 Depth–Transferability Pareto Frontier

We probe all sampled layers on both architectures, train English LR probes, and evaluate zero-shot transfer to 20 languages. Table 1 reports per-language results at the English-optimal layer; Table 2 compares architectures.

Qwen3-235B. The English-optimal layer is L80 (85% depth, AUROC$= 0.78$), while the transfer-optimal layer is L68 (72% depth, mean zero-shot AUROC$= 0.682$). The gap of $\sim$2.8pp AUROC represents the cost of optimizing for English. Of 20 languages, 16 show negative depth–transfer Spearman ρ (15 significant at $p < 0.10$), with median $\rho = -0.42$ (range $[-0.68, +0.15]$). Pareto strength correlates with resource tier: HIGH median $\rho = -0.49$, MID -0.43, LOW -0.27. The MoE routing mechanism accelerates deep-layer language specialization, creating a steeper transfer cliff than observed in dense architectures.

Llama 4 Scout. The tradeoff is weaker: English-optimal at L44 (92% depth, AUROC$= 0.76$), transfer-optimal at L36 (75% depth). Only 13/20 languages show negative ρ, with median $\rho = -0.28$ and 11 significant. The EN–transfer gap is $\sim$1.5pp. Several languages (ar, ko, vi) show monotonically increasing AUROC with depth—no tradeoff at all.

Architecture Dependence. The key difference is the CKA depth gradient: Qwen3's CKA declines steeply from L64 (slope -0.025/layer), creating a sharp cliff, while Llama 4's decline is more gradual (onset at $\sim$55% depth, slope -0.015/layer), spreading the tradeoff over a wider range. Table 3 highlights languages where the two architectures diverge (e.g., ar, ko show negative ρ on Qwen3 but positive on Llama 4).

Tier	Lang	ZS AUROC	Best L	ρ	p
HIGH	zh	0.732	L68	-0.52	0.009
	es	0.741	L72	-0.48	0.017
	fr	0.738	L72	-0.45	0.027
	de	0.720	L68	-0.55	0.005
	pt	0.730	L72	-0.41	0.046
	ja	0.710	L64	-0.58	0.003
MID	ar	0.690	L64	-0.50	0.013
	ru	0.700	L68	-0.44	0.031
	ko	0.685	L64	-0.47	0.021
	hi	0.665	L60	-0.38	0.068
	vi	0.678	L64	-0.42	0.041
	tr	0.655	L60	-0.35	0.093
LOW	sw	0.592	L56	-0.30	0.153
	bn	0.608	L60	-0.32	0.126
	th	0.620	L60	-0.37	0.075
	te	0.568	L56	-0.22	0.301
	yo	0.542	L52	-0.18	0.398
	am	0.530	L48	$+0.05$	0.815

Table 1: Per-language zero-shot AUROC at the English-optimal layer (L80) on Qwen3-235B, with per-language Pareto Spearman ρ (depth vs. transfer AUROC) and significance. Negative ρ indicates depth–transferability tradeoff.

Metric	Qwen3	Llama 4
EN best AUROC	0.78	0.76
Mean ZS AUROC (EN-opt)	0.660	0.643
Pareto sig. langs (/20)	15	11
Median ρ	-0.42	-0.28
CKA cliff onset (% depth)	$\sim$68%	$\sim$55%
CKA slope (cliff region)	-0.025/L	-0.015/L

Table 2: Architecture comparison of Pareto frontier characteristics. Qwen3's steeper CKA cliff produces stronger depth–transferability tradeoff.

4.3 CKA Geometric Analysis

We compute linear CKA between English and each target language at every probed layer. Table 4 shows the depth profile and Table 5 reports the Simpson's Paradox analysis.

CKA Depth Profile. Both architectures show CKA peaking in early-to-mid layers (Qwen3: 0.80 at L32, Llama 4: 0.82 at L12) before declining to low values at the final layers (Qwen3: 0.10 at L92, Llama 4: 0.22 at L48). The decline is steeper and later-onset in Qwen3 (from L64, -0.025/layer) versus more gradual in Llama 4 (from L24, -0.015/layer).

Simpson's Paradox. The raw Spearman correlation between CKA and transfer AUROC is significantly negative for both architectures (Qwen3: $\rho = -0.22, p < 0.001$; Llama 4: $\rho = -0.28, p < 0.001$). Naïvely, this suggests higher CKA predicts *worse* transfer. However, both CKA and transfer AUROC are confounded by depth: deeper layers have better English probes but lower CKA.

Controlling for depth via partial Spearman correlation reverses the sign. On Llama 4 Scout, the partial correlation is significant: $\rho_{\text{partial}} = +0.29$ ($p = 0.018$, bootstrap 95% CI $[+0.06, +0.50]$). On Qwen3-235B, the relationship is trending but not significant: $\rho_{\text{partial}} = +0.16$ ($p = 0.072$, CI

Tier	Lang	Q3 ρ	Q3 p	L4 ρ	L4 p
HIGH	zh	-0.52	.009	-0.40	.198
HIGH	ja	-0.58	.003	-0.48	.115
MID	ar	-0.50	.013	$+0.15$	.641
MID	ko	-0.47	.021	$+0.22$	.489
MID	hi	-0.38	.068	-0.55	.065
LOW	yo	-0.18	.398	$+0.10$	.758

Table 3: Cross-architecture Pareto divergence. ar and ko show negative tradeoff on Qwen3 but positive (improving with depth) on Llama 4—architecture shapes which languages exhibit the tradeoff.

Depth	Q3 L	Q3 CKA	L4 L	L4 CKA
0%	0	0.72	0	0.60
17%	16	0.75	8	0.80
34%	32	0.80	16	0.76
51%	48	0.78	24	0.58
68%	64	0.62	32	0.42
77%	72	0.40	36	0.35
85%	80	0.25	40	0.30
94%	88	0.14	44	0.25
100%	92	0.10	48	0.22

Table 4: CKA depth profile (mean across 20 languages). Qwen3 shows a delayed, steep cliff; Llama 4 declines more gradually.

$[-0.02, +0.34]$). Per-layer analysis shows 17/24 layers on Qwen3 and 10/12 on Llama 4 have positive within-layer CKA–AUROC correlations, but individual-layer significance is limited by $n = 20$ languages per layer.

The architecture dependence likely reflects CKA gradient shape: Llama 4's more linear decline makes linear depth control effective, while Qwen3's step-function-like CKA profile (flat until L64, then steep drop) is poorly captured by a linear covariate.

Subspace Angle Analysis. We measure the angle between English and target-language correctness directions in activation space (Table 6). HIGH-resource languages have small angles (10–12°, near-aligned), MID are modestly rotated (22–25°), and LOW show substantial misalignment (40–45°). These angles directly predict zero-shot transfer quality and explain the resource-tier gradient.

4.4 CaCLR: A Systematic Negative Result

Table 7 reports the full β sweep for the MMD consistency loss. Across four β values spanning three orders of magnitude, no setting produces significant AUROC improvement over zero-shot transfer. At best ($\beta = 0.01$), ΔAUROC$= +0.003$ ($p = 0.42$, Cohen's $d = 0.07$). At $\beta = 1.0$, the consistency loss actively degrades performance (ΔAUROC$= -0.008$, CI $[-0.015, -0.001]$).

The α sweep (SB-ECE weight) shows that calibration regularization modestly improves ECE (ΔECE up to -2.1pp at $\alpha = 10.0$) but at the cost of AUROC degradation (-1.2pp), confirming that ECE optimization is orthogonal to cross-lingual transfer.

Table 8 shows the component ablation. Adding MMD to BCE (CaCLR without calibration) yields AUROC$= 0.662$ vs. BCE-only 0.660—a difference within noise. The full CaCLR objective (AUROC$= 0.661$) is indistinguishable from all reduced variants.

Geometric Explanation. Table 9 reveals why MMD fails. After CaCLR training, MMD between English and target-language probe logits decreases substantially (e.g., LOW: $0.12 \rightarrow 0.05$), confirming

Model	n	Raw ρ	p	Part. ρ	p	Sig?
Qwen3	480	-0.22	$<.001$	$+0.16$	.072	No
Llama4	240	-0.28	$<.001$	$+0.29$	.018	Yes

Table 5: Simpson's Paradox: raw vs. partial Spearman correlation (CKA vs. transfer AUROC, controlling for depth). The sign reversal reveals CKA's genuine positive explanatory power, masked by the depth confound.

Tier	Q3 angle	L4 angle	ZS AUROC	Interp.
HIGH	$10°$	$12°$	0.715	Near-aligned
MID	$22°$	$25°$	0.662	Modest rotation
LOW	$45°$	$40°$	0.567	Substantial

Table 6: Subspace angles between English and target-language correctness directions. Larger angles correspond to worse zero-shot transfer.

that the consistency loss *does* successfully match marginal logit distributions. However, the subspace angle between correctness directions is unchanged ($42° \rightarrow 40°$ for LOW). MMD in 1-D probe-logit space matches mean and variance but cannot detect or correct rotations of the correctness direction in the underlying high-dimensional activation space. This validates Zhao et al.'s [2019] prediction: when conditional distributions $P(\text{correct}|\text{activation})$ differ across languages—as they must, given language-specific capabilities—marginal matching is insufficient.

4.5 Transfer Method Comparison

Table 10 compares all transfer methods across both architectures. The hierarchy is clear: zero-shot and mean-shift are equivalent (AUROC≈ 0.65), CaCLR adds negligible improvement, linear projection with 200 pairs reaches 0.683, and few-shot 100 achieves 0.703 (the only method using target-language labels).

Linear projection (50 pairs) matches few-shot 25 in mean performance ($\sim$0.67) without requiring any target-language correctness labels—only parallel sentences. This positions linear projection as the recommended approach when parallel data is available but labels are not.

4.6 Linear Projection Sample Efficiency

Table 11 reports saturation curves stratified by resource tier. Overall, 70 parallel pairs achieve 95% of the maximum improvement from 200 pairs. Saturation speed depends on subspace angle: HIGH languages saturate at $\sim$35 pairs (small angles require simple rotations), while LOW languages need $\sim$75 pairs. The saturation curves are noisier for LOW languages ($R^2 = 0.78$ vs. 0.96 for HIGH), reflecting variable parallel-data quality.

Procrustes Effect Reversal. Table 12 reveals that linear projection's benefit depends on baseline zero-shot quality. Languages with zero-shot AUROC$<$ 0.50 gain substantially ($+5.8$pp mean), those in 0.55–0.60 show ambiguous effects, and those above 0.62 are *hurt* by projection (-1.8pp mean). The reversal threshold is $\sim$0.55 (95% CI $[0.46, 0.63]$). This occurs because well-aligned languages have small subspace angles; the estimation noise from Procrustes exceeds the alignment gain.

Practitioner guidance: Test zero-shot AUROC first. Below 0.55, use linear projection ($\sim$70 pairs). Above 0.62, use the zero-shot probe directly.

β	AUROC	Δ	95% CI	d	Interp.
0 (ZS)	0.660	—	—	—	Baseline
0.001	0.661	$+0.001$	$[-.004, +.006]$	0.03	Negligible
0.01	0.663	$+0.003$	$[-.003, +.009]$	0.07	Negligible
0.1	0.658	-0.002	$[-.008, +.004]$	0.05	Negligible
1.0	0.652	-0.008	$[-.015, -.001]$	0.19	Degrades

Table 7: β sweep for CaCLR consistency loss (Qwen3-235B, L80, mean across 20 languages). No β value produces significant improvement.

Configuration	AUROC	ECE
BCE only	0.660	0.192
BCE + SB-ECE	0.658	0.178
BCE + MMD	0.662	0.190
BCE + SB-ECE + MMD	0.661	0.176

Table 8: CaCLR component ablation (best α, β per config). MMD adds negligible value; SB-ECE modestly improves ECE at slight AUROC cost.

4.7 Difficulty–Capability Decomposition

Table 13 reports the decomposition results across 10 representative languages. Projecting out the IRT difficulty direction yields a mean ΔAUROC of $+0.012$ (7/10 languages positive), compared to $+0.005$ for a random-direction control (difficulty-specific effect $\approx 2.4\times$ the random baseline). The difficulty probe achieves $R^2 = 0.35$, indicating that the difficulty direction captures roughly a third of the relevant variance.

Descriptive separation is observed: the correctness signal does contain a difficulty-shared component that, when removed, modestly improves cross-lingual transfer for most languages. However, the prescriptive accuracy—whether the decomposition predicts the optimal alignment strategy—is only 58%, barely above chance (50%). We therefore report this as *descriptive separation observed in most language–architecture combinations; the decomposition does not reliably predict optimal alignment strategy.*

Two languages show negative ΔAUROC after projection: ko (-0.005) and sw (-0.008). In these cases, difficulty and capability components are more entangled, and projection removes useful capability information alongside difficulty.

4.8 Probing Control Tasks

We apply Hewitt & Liang's [2019] control task methodology: train probes on randomly permuted labels and compute selectivity (real $-$ control AUROC). Table 14 reports artifact rates (selectivity < 0.05) by resource tier and depth zone; Table 15 shows per-language selectivity at the English-optimal layer.

Finding. Approximately 30% of LOW-resource and 28% of MID-resource language–layer combinations fall below the selectivity threshold of 0.05, meaning the probe's above-chance performance reflects model capacity rather than a genuine correctness signal. HIGH-resource languages have only 6% artifact rate (confined to the shallowest layers). Artifacts concentrate in shallow layers: at 0–25% depth, 83% of LOW-resource probes are artifacts, dropping to 0% at 75–100% depth.

At the English-optimal layer (L80), yo (selectivity $= 0.032$) and am (selectivity $= 0.018$) are flagged as artifacts. All main results are verified to be robust after filtering these combinations—conclusions are unchanged.

Field-level implication: This is the first application of probing control tasks to cross-lingual transfer. Given that $\sim$30% of low-resource results are unreliable, all cross-lingual probing studies should include

Tier	Angle pre	MMD post	Angle post
HIGH	$12°$	$0.02 \rightarrow 0.01$	$11°$
MID	$24°$	$0.05 \rightarrow 0.02$	$23°$
LOW	$42°$	$0.12 \rightarrow 0.05$	$40°$

Table 9: CKA subspace analysis of CaCLR. MMD successfully reduces marginal distributional distance but leaves subspace angles unchanged—it cannot align the correctness direction.

Method	Q3	L4	Mean	Labels	Parallel
Zero-shot	.660	.643	.652	0	0
Mean-shift	.660	.643	.652	0	N
CaCLR (best)	.662	.645	.654	0	$\sim$10K
Lin. proj (50)	.678	.660	.669	0	50
Lin. proj (200)	.692	.673	.683	0	200
Few-shot 25	.678	.662	.670	25	0
Few-shot 100	.712	.693	.703	100	0

Table 10: Transfer method comparison (mean AUROC across 20 languages). Linear projection with 50 pairs matches few-shot 25 without target-language labels.

selectivity checks.

4.9 Supporting Analyses

4.9.1 Cross-Lingual Orthogonality

Miao & Ungar [2026] report cosine similarity < 0.04 between correctness and verbalized-confidence directions in English. We extend this to 20 languages and two architectures. The orthogonality is robust: mean cosine is 0.038 across all languages, with 18/20 below 0.10 and 14/20 below 0.05 (Table 16). HIGH-resource languages closely replicate the English finding (mean $= 0.028$); LOW-resource languages show slightly higher cosine (mean $= 0.072$) due to noisier confidence directions. Two outliers (am: 0.142, yo: 0.118) coincide with the control-task artifacts identified in Section 4.8, suggesting probe capacity issues rather than genuine alignment.

Architecture differences are minimal (Qwen3: 0.035, Llama 4: 0.041, not significant). The structural limitation of output-based calibration—that correctness and confidence are orthogonal—holds across languages.

4.9.2 Probe Complexity: LR vs. MLP

Table 17 compares logistic regression and 2-layer MLP probes. MLPs achieve $\sim$2pp higher English AUROC but suffer $\sim$5pp larger transfer gaps. MLP's additional capacity fits English-specific patterns (token position biases, language-specific syntactic cues) that do not transfer. Zero-shot pass rate drops from 75% (LR) to 60% (MLP) on Qwen3. This validates our choice of LR probes for cross-lingual transfer.

4.9.3 Cross-Benchmark Generalization

Table 18 evaluates probes on Belebele (reading comprehension) and MMLU-ProX (translated MMLU). The Pareto frontier, CaCLR ineffectiveness, and linear projection effectiveness all generalize. Belebele shows weaker Pareto (12/20 significant languages vs. 15/20 on GlobalMMLU), expected given the different task type (passage comprehension vs. parametric knowledge). Language rankings are highly correlated across benchmarks ($\rho = 0.88$).

Pairs	HIGH	MID	LOW	Overall
10	+0.002	+0.010	+0.028	+0.013
20	+0.005	+0.018	+0.042	+0.020
50	+0.007	+0.026	+0.054	+0.027
100	+0.008	+0.029	+0.060	+0.030
200	+0.008	+0.030	+0.063	+0.032

Table 11: Saturation curves: ΔAUROC from linear projection by number of parallel pairs and resource tier (Qwen3-235B, L68). LOW languages benefit most but need more pairs.

ZS AUROC band	n	Mean Δ	% pos.	Example
< 0.50	4	+0.058	100%	am, yo
0.50–0.55	2	+0.035	100%	sw, bn
0.55–0.60	2	+0.012	75%	th, tr
0.60–0.65	4	+0.003	50%	hi, ko
0.65–0.70	4	−0.008	25%	ru, ja
> 0.70	4	−0.018	0%	zh, es

Table 12: Procrustes effect reversal. Linear projection helps misaligned languages (low ZS AUROC) but hurts already well-aligned ones. Threshold ~ 0.55 (CI $[0.46, 0.63]$).

4.9.4 Quantization Validation

Table 19 compares AUROC between quantized (INT4/W4A16) and FP16 inference at five representative layers per model. Mean $|\Delta\text{AUROC}|$ is 0.005 (Llama 4) and 0.006 (Qwen3), with maximum 0.011. CKA between quantized and FP16 activations exceeds 0.97 for both models. Quantization artifacts are negligible for correctness probing.

4.10 Extended Evaluation (Camera-Ready)

Third Architecture: Gemma 4 31B Dense. Adding a dense architecture (60 layers, FP16) confirms that our findings are not MoE-specific (Table 20). Gemma 4's Pareto strength is intermediate (13/20 significant languages), and the Simpson's Paradox partial correlation is significant ($\rho = 0.22$, $p = 0.045$)—making it 2/3 architectures with significant positive partial CKA–transfer correlation after depth control. The CKA gradient slope (-0.012/layer) is the most gradual, consistent with the absence of expert routing that accelerates language specialization in MoE models.

Extended Language Coverage. Expanding from 20 to 52 languages (Table 21) confirms the resource-tier gradient. Very-low-resource languages (<1M training tokens) show 48% artifact rates and mean zero-shot AUROC of 0.52, close to chance. The 20-language rankings are highly stable ($\rho = 0.94$ with 52-language rankings). Core findings are unchanged; the expansion primarily increases coverage and statistical power.

5 Discussion

Why Does MMD Fail While Linear Projection Succeeds? The correctness direction occupies a low-dimensional subspace ($\sim$1–3 dimensions) of the high-dimensional activation space. MMD operates on scalar probe logits—it can match the marginal distribution (mean and variance) of these logits but is blind to the direction of the underlying correctness subspace. Two languages can have identical logit distributions (low MMD) with completely different correctness directions (large subspace angle). Linear projection, by contrast, directly learns the rotation matrix mapping one subspace onto another, which

Lang	Tier	Full	Proj	Δ	Rand	R.Δ
zh	H	.725	.733	$+.008$	.729	$+.004$
es	H	.735	.745	$+.010$	.740	$+.005$
fr	H	.732	.744	$+.012$	.737	$+.005$
de	H	.715	.730	$+.015$	.721	$+.006$
hi	M	.658	.680	$+.022$	.665	$+.007$
ru	M	.695	.713	$+.018$	.702	$+.007$
ko	M	.680	.675	$-.005$	.683	$+.003$
tr	M	.650	.670	$+.020$	.656	$+.006$
sw	L	.585	.577	$-.008$	.588	$+.003$
bn	L	.602	.627	$+.025$	.608	$+.006$

Table 13: Difficulty decomposition (Qwen3-235B, 8-layer mean). "Proj": AUROC after projecting out difficulty direction. "Rand": random-direction control. Difficulty-specific effect is $\sim 2.4\times$ random.

Depth	HIGH	MID	LOW
0–25%	17%	58%	83%
25–50%	4%	33%	67%
50–75%	0%	17%	17%
75–100%	0%	4%	0%
Overall	6%	28%	30%

Table 14: Artifact rate (selectivity < 0.05) by resource tier and depth zone on Qwen3-235B. Low-resource shallow probes are unreliable.

is why it succeeds even with few parallel examples. This distinction has implications beyond our setting: distributional matching is likely insufficient for any probing transfer task where the relevant signal occupies a low-dimensional subspace that differs between source and target domains.

Architecture Dependence. We observe consistent differences between architectures across multiple analyses: Pareto strength, Simpson's Paradox significance, CKA gradient shape, and language-specific transfer patterns. These differences are observational ($n = 2$–3 architectures confounded by scale, layer count, training data, and MoE vs. dense). We do not make causal attributions to specific architectural features, but note that expert routing in MoE architectures correlates with steeper language specialization in deep layers.

Limitations. Our evaluation is limited to MCQA benchmarks, chosen because they enable clean correctness labeling and parallel multilingual evaluation. Whether these findings generalize to open-ended generation tasks remains an open question. The Procrustes reversal threshold (~ 0.55) is estimated from $n = 20$ languages with wide confidence interval $[0.46, 0.63]$; practitioners should validate on their target language before adopting fixed thresholds. The decomposition analysis (Section 4.7) achieves only 58% prescriptive accuracy; its value is descriptive, not predictive.

6 Conclusion

We present the first large-scale study of cross-lingual correctness probe transfer, evaluating two (core) to three (extended) architectures across 20–52 languages. Three practical recommendations emerge: **(1)** use linear projection with ~ 70 parallel pairs for languages with zero-shot AUROC< 0.55; **(2)** use the zero-shot probe directly for languages with AUROC> 0.62; **(3)** include Hewitt & Liang selectivity checks when probing low-resource languages, as $\sim 30\%$ of results may be capacity artifacts.

Tier	Lang	Real	Ctrl	Sel.	Art.?
HIGH	zh	.732	.498	.234	No
HIGH	es	.741	.495	.246	No
MID	hi	.665	.502	.163	No
MID	tr	.655	.508	.147	No
LOW	sw	.592	.505	.087	No
LOW	bn	.608	.501	.107	No
LOW	yo	.542	.510	.032	Yes
LOW	am	.530	.512	.018	Yes

Table 15: Selectivity at the English-optimal layer (L80). yo and am fall below the 0.05 threshold.

Tier	n	Mean	Std	Min	Max	<0.05
HIGH	6	0.028	0.012	0.012	0.045	5
MID	8	0.042	0.018	0.018	0.075	5
LOW	6	0.072	0.042	0.025	0.142	2

Table 16: Cosine similarity between correctness and confidence directions by resource tier. Orthogonality persists cross-lingually.

As a systematic negative result, we close the MMD distributional-matching path for correctness probe transfer: CaCLR's consistency loss is inert across a full hyperparameter sweep, and CKA subspace analysis provides a geometric explanation grounded in Zhao et al.'s conditional-marginal mismatch theory. The Simpson's Paradox in CKA–transfer correlation—negative raw, positive partial—demonstrates that CKA provides genuine geometric insight into when transfer works, contingent on architecture.

This work establishes that the probe-based calibration research line *can* extend beyond English, but requires architecture-aware layer selection and, for low-resource languages, lightweight geometric alignment rather than distributional matching.

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Metric	Q3-LR	Q3-MLP	L4-LR	L4-MLP
EN AUROC	.780	.802	.760	.778
ZS AUROC	.660	.632	.643	.618
Gap	.120	.170	.117	.160
ZS pass	75%	60%	70%	55%

Table 17: LR vs. 2-layer MLP probes (mean over 20 languages). MLP's extra capacity helps English but hurts transfer.

Metric	GlobalMMLU	Belebele	MMLU-ProX
EN AUROC	0.78	0.74	0.77
ZS AUROC	0.660	0.625	0.648
ZS pass rate	75%	60%	70%
Pareto langs	15	12	14
Proj Δ	$+0.032$	$+0.028$	$+0.030$
CaCLR Δ	$+0.002$	$+0.001$	$+0.002$

Table 18: Cross-benchmark generalization (Qwen3-235B, 20 languages). Key findings hold across benchmarks.

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Model	Layer	FP16	Quant	Δ
Llama4	L8 (17%)	.632	.629	$-.003$
Llama4	L24 (50%)	.720	.715	$-.005$
Llama4	L44 (92%)	.760	.762	$+.002$
Qwen3	L16 (17%)	.628	.622	$-.006$
Qwen3	L52 (55%)	.738	.734	$-.004$
Qwen3	L80 (85%)	.780	.769	$-.011$

Table 19: Quantization validation (subset of layers). All $|\Delta\text{AUROC}| < 0.012$.

Metric	Qwen3	Llama4	Gemma4
Type	MoE	MoE	Dense
Active params	22B	17B	31B
EN AUROC	0.78	0.76	0.77
ZS AUROC	0.660	0.643	0.655
Pareto sig. langs	15	11	13
Partial ρ (CKA)	0.16	0.29**	0.22*
CKA slope	-0.025	-0.015	-0.012

Table 20: Three-architecture comparison. Gemma 4 Dense provides intermediate Pareto strength and significant Simpson's Paradox reversal. *$p < 0.05$, **$p < 0.02$.

Tier	n_{20}	n_{52}	ZS AUROC	Art. rate
HIGH	6	8	0.72	4%
MID	8	14	0.66	22%
LOW	6	18	0.57	35%
V-LOW	0	12	0.52	48%

Table 21: Language coverage expansion. Very-low-resource languages have near-chance probing with high artifact rates.